

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2180

Canon City 138 kV
December 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2180. The Interconnection Customer proposes a 149.0 MW battery energy storage facility interconnecting at the Canon City 138 kV substation in the Virginia area. The request was received on September 09, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 652 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$124,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Centralia – Cedar Bayou Plant 138 KV Line is overloaded to 119.2% of its 303 MVA emergency rating for the loss of the Pheasant Branch – Switzer 138 KV. The project contributes 5.2% DFAX (7.7 MW) to this overload.

For the loss of Pheasant Branch – Switzer 138 KV, the Cedar Bayou Plant Substation – Jim Hill 138 KV L loads to 117.6% of its 352 MVA rating. Project DFAX: 5.2%.

For the loss of Switzer – Pheasant Branch Substation 138 KV line, the Haggin – Jim Hill 138 KV loads to 116.1% of its 413 MVA rating. Project DFAX: 5.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cedar Bayou Plant Substation – Centralia 138 KV L	Rebuild line with higher-capacity conductor	\$1,184,000	2% (\$25,000)
Cedar Bayou Plant Substation – Jim Hill 138 KV L	Rebuild line with higher-capacity conductor	\$3,700,000	2% (\$79,000)
Jim Hill – Hagin 138 KV	Rebuild line with higher-capacity conductor	\$944,000	2% (\$20,000)
	Total allocated to project		\$124,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.